



BUS RESERVATION SYSTEM

Section: 3W8



Instructor's name: Salha Alasmari

Student Name:	Student ID:
Shujon Ali ALshehri	445008413
Fair Omar Alrubeaan	445008461
Remas Alqaahtani	445008484
Leena Alajlan	445008525

Overview Project

This report focuses on creating a Bus Reservation System using assembly language in EMU8086. It tackles the main issue of letting users check out bus routes and book tickets. We'll share our approach to designing the program, highlighting its key features like choosing a bus and figuring out the total cost. The report will also give an overview of the code and how it all works.

1-The Project description

The Bus Reservation System project is designed to meet the increasing demand for an efficient and user-friendly application that enables passengers to view available bus routes and make reservations seamlessly. As travel options continue to expand, there is a critical need for streamlined booking systems that deliver essential information directly to users. This application aims to enhance the travel experience by providing an intuitive interface for easy navigation and informed decision-making.

2-The Problem Solution

This application aims to enhance the travel experience by providing an intuitive interface for easy navigation and informed decision-making.

4-Screen of code

```

0001 .model small
0002 .stack 100h
0003 .data
0004
0005 menu db 0dh,0ah,0dh,0ah,' BUS RESERVATION SYSTEM ',0dh,0ah,0dh,0ah
0006 db ' 1- Displaying all available Buses and their details ',0dh,0ah
0007 db ' 2- Booking a Bus ',0dh,0ah
0008 db ' 3- Calculating Total Price ',0dh,0ah
0009 db ' 4- Exit the application ',0dh,0ah,'$'
0010
0011 msg db 0dh,0ah,0dh,0ah,' Please select your option: $',0dh,0ah
0012
0013 bus_sch db 0dh,0ah,0dh,0ah,' Buses schedule ',0dh,0ah
0014 db ' 1 Bus Number 1 Departure Destination 1 Departure Time 1 Ticket Price:',0dh,0ah
0015 db ' 1 Bus 101 1 Riyadh - Jeddah 1 9:00 AM 1 100 SAR:',0dh,0ah
0016 db ' 1 Bus 102 1 Riyadh - Dammam 1 11:00 PM 1 120 SAR:',0dh,0ah
0017 db ' 1 Bus 103 1 Jeddah - Riyadh 1 21:00 PM 1 90 SAR:',0dh,0ah,'$'
0018
0019 apo db 0dh,0ah,0dh,0ah,' Sorry, not Available if . press any character to try again $',0dh,0ah
0020 str db 0dh,0ah,0dh,0ah,' Are you a student? if you are Press "S" to get 30% discount. if not press any character $',0dh,0ah
0021 noReserve db 0dh,0ah,0dh,0ah,' Sorry, you dont have any booking yet. press any character to return to the menu $',0dh,0ah
0022 total_msg db 0dh,0ah,0dh,0ah,' The total price is: $',0dh,0ah
0023 total_msg0 db 0dh,0ah,0dh,0ah,' The total price after discount is: $',0dh,0ah
0024 end_msg db 0dh,0ah,0dh,0ah,' Thank you for using our bus reservation system, have a nice day <3 $'
0025
0026 n70 du 70
0027 ni00 du 100
0028 total du ?
0029

```

-The START label is the entry point, which loads and displays the main menu

```

0035 .code
0036 main proc
0037 mov ax,0data
0038 mov ds,ax
0039
0040
0041 START:
0042 mov ah,9
0043 lea dx,menu
0044 int 21h
0045
0046 mov ah,9
0047 lea dx,msg
0048 int 21h
0049
0050 mov ah,1
0051 int 21h
0052
0053 cmp al,"1"
0054 jl out_of_menu
0055 cmp al,"4"
0056 jg out_of_menu
0057
0058 cmp al,"1"
0059 je show_sch
0060
0061 cmp al,"2"
0062 je reserve
0063
0064 cmp al,"3"
0065 je calculate
0066
0067 cmp al,"4"
0068 je exit
0069

```

-The OUT_OF_MENU label might handle cases where an invalid option is selected, directing the user back to the main menu.

- The SHOW label displays the bus schedule.

```

0069
0070 OUT_OF_MENU:
0071 mov ah,9
0072 lea dx,apo
0073 int 21h
0074
0075 mov ah,1
0076 int 21h
0077 jmp start
0078
0079 SHOW_SCH:
0080 mov ah,9
0081 lea dx,bus_sch
0082 int 21h
0083 jmp start
0084
0085
0086
0087 RESERVE:
0088 mov ah,9
0089 lea dx,bus_sch
0090 int 21h
0091
0092 mov ah,9
0093 lea dx,msg
0094 int 21h
0095
0096 call indec
0097 mov bx,ax
0098
0099 cmp bx,101
0100 jl out_of_sch
0101 cmp bx,103
0102 jg out_of_sch
0103 jmp start
0104
0105 OUT_OF_SCH:
0106 mov ah,9
0107 lea dx,apo
0108 int 21h
0109
0110 mov ah,1
0111 int 21h
0112 jmp reserve
0113
0114
0115

```

-THE RESERVE label displays the bus schedule and ask the user to choose the bus number that suits the user.

-The OUT_Of_SCH label Similar to OUT_OF_MENU, handles invalid inputs specifically for the schedule display, directing back to the reservation process.

- The CALCULATE label compares user input to determine the ticket price based on conditions
- The NO_RESERVE label If no reservation is made and the user wants to calculate the price, a message is displayed, and the program returns to the start.
- The BILL label displaying the total price. It checks for a discount option based on user input.

```

115
116 CALCULATE:
117 cmp bx,101
118 je price_100
119
120 cmp bx,102
121 je price_120
122
123 cmp bx,103
124 je price_90
125
126 jmp no_reserve
127
128
129 PRICE_100:
130 mov total,100
131 jmp bill
132
133 PRICE_120:
134 mov total,120
135 jmp bill
136
137 PRICE_90:
138 mov total,90
139 jmp bill
140
141
142 NO_RESERVE:
143 mov ah,9
144 lea dx,noReserve
145 int 21h
146
147 mov ah,1
148 int 21h
149 jmp start
150
151
152 BILL:
153 mov ah,9
154 lea dx,stu
155 int 21h
156
157 mov ah,1
158 int 21h
159
160 cmp al,"S"
161 je dis_30
162 jne no_dis
163

```

- The DIS_30 label Applies a 30% discount to the total if the user is student and displaying the discounted total.
- The NO_DIS label Displays the total without any discount.
- The EXIT label Ends the program by displaying an exit message.

```

164
165 DIS_30:
166 mov ax,total
167 mul n70
168 div n100
169 mov total,ax
170
171 mov ah,9
172 lea dx,total_msgD
173 int 21h
174
175 mov ax,total
176 call outdec
177
178 jmp start
179
180
181 NO_DIS:
182 mov ah,9
183 lea dx,total_msg
184 int 21h
185
186 mov ax,total
187 call outdec
188
189 jmp start
190
191
192 EXIT:
193 mov ah,9
194 lea dx,end_msg
195 int 21h
196
197 mov ah,4ch
198 int 21h
199
200 main endp
201 include C:\enu8086\MySource\i1dec.asm
202 include C:\enu8086\MySource\outdec.asm
203 end

```

5-Sample Output:

Select option 1

```
emulator screen (106x38 chars)
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application

Please select your option: 1

Buses schedule
-----
| Bus Number | Departure Destination | Departure Time | Ticket Price |
-----
| Bus 101    | Riyadh - Jeddah      | 9:00 AM        | 100 SAR       |
-----
| Bus 102    | Riyadh - Dammam      | 11:00 PM       | 120 SAR       |
-----
```

Select option 2

-Select available Bus in Buses schedule.

```
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application

Please select your option: 2

Buses schedule
-----
| Bus Number | Departure Destination | Departure Time | Ticket Price |
-----
| Bus 101    | Riyadh - Jeddah      | 9:00 AM        | 100 SAR       |
-----
| Bus 102    | Riyadh - Dammam      | 11:00 PM       | 120 SAR       |
-----
| Bus 103    | Jeddah - Riyadh      | 21:00 PM       | 90 SAR        |
-----

Please select your option: :103
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application
```

- Select NOT available Bus in Buses schedule.

```
emulator screen (106x38 chars)

Please select your option: 2

Buses schedule
-----
| Bus Number | Departure Destination | Departure Time | Ticket Price |
-----
| Bus 101    | Riyadh - Jeddah      | 9:00 AM        | 100 SAR       |
-----
| Bus 102    | Riyadh - Dammam      | 11:00 PM       | 120 SAR       |
-----
| Bus 103    | Jeddah - Riyadh      | 21:00 PM       | 90 SAR        |
-----

Please select your option: :104
Sorry, not Available :( , press any character to try again

Buses schedule
-----
| Bus Number | Departure Destination | Departure Time | Ticket Price |
-----
| Bus 101    | Riyadh - Jeddah      | 9:00 AM        | 100 SAR       |
-----
| Bus 102    | Riyadh - Dammam      | 11:00 PM       | 120 SAR       |
-----
| Bus 103    | Jeddah - Riyadh      | 21:00 PM       | 90 SAR        |
-----
```

Select option 3

- if there is no booking to calculate.

```
emulator screen (100x33 chars)

BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application

Please select your option: 3
sorry, you dont have any booking yet
```

- calculate price with discount.

```
emulator screen (100x33 chars)

| Bus 102 | Riyadh - Dammam | 11:00 PM | 120 SAR |
| Bus 103 | Jeddah - Riyadh | 21:00 PM | 90 SAR |

Please select your option: :102
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application

Please select your option: 3
Are you a student? if you are Press "S" to get 30% discount, if not press any character d
The total price is: 120
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application
```

- calculate price without discount.

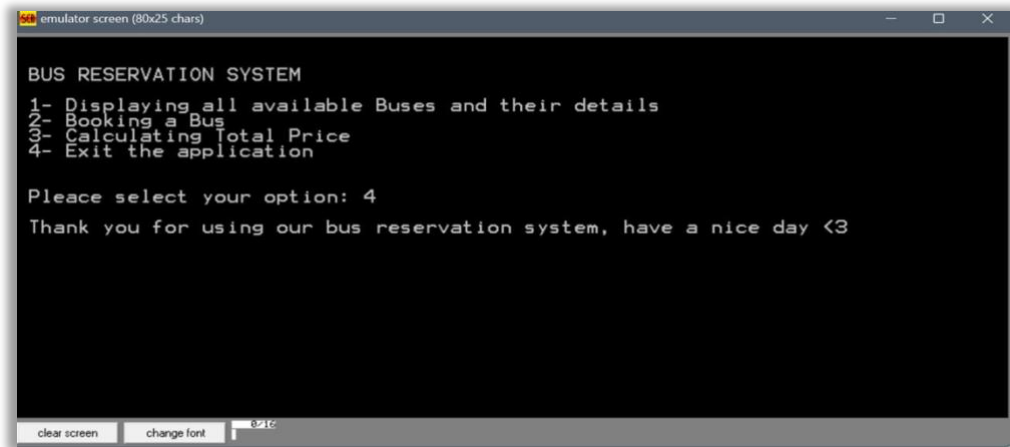
```
emulator screen (100x33 chars)

| Bus 102 | Riyadh - Dammam | 11:00 PM | 120 SAR |
| Bus 103 | Jeddah - Riyadh | 21:00 PM | 90 SAR |

Please select your option: :102
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application

Please select your option: 3
Are you a student? if you are Press "S" to get 30% discount, if not press any character s
The total price after discount is: 84
BUS RESERVATION SYSTEM
1- Displaying all available Buses and their details
2- Booking a Bus
3- Calculating Total Price
4- Exit the application
```

Select option 4



Select option 5

